thin mica window and the center wire is terminated with a glass bead. The cathode is always at ground potential while the anode is at a high positive potential.

When the radiation intensity is measured as a function of the electrode potential it is found for GM counters (as for proportional counters) that over a certain voltage interval there is little, if any, change in the measured count rate. This is known as the *plateau region* (Fig. 9.9). This is the voltage range in which the detectors are operated since they are relatively insensitive to small voltage changes in this region. For organic quenched GM tubes the plateau commonly occurs between 1200 and 1500 V and its slope should not exceed 5% per 100 V. Halogen-quenched tubes operate at a lower voltage but have higher plateau slopes (cf. Table 9.2).

9.4. SEMICONDUCTOR DETECTORS

The semiconductor detector is similar to an ordinary semiconductor diode composed of p-type and n-type semiconductor material. This detector has become dominant for nuclear spectroscopy (i.e. determination of the energy of nuclear radiation) but it is not so often used for simple measurement of count rates.

Semiconductors are materials like silicon (resistivity ~10^3 Ωm) and germanium (0.6 Ωm) with resistivities between metals (e.g. copper, 10^{-8} Ωm) and insulators (e.g. quartz, 10^{12} Ωm). A crystal of pure silicon placed between two electrodes is almost nonconducting. The electrons in the material are almost all *valence electrons*, bound to specific silicon atoms with an energy of 1.115 eV at room temperature (0.75 eV for Ge at 80 K). If 1.115 eV is given to an electron in silicon, it moves to a band of overlapping energy levels which are not associated with specific atoms. The electron moves readily through the crystal in this "conduction band", i.e. the crystal conducts electricity. At a certain temperature some electrons, according to the Maxwell energy distribution, always have the necessary energy to be in the conduction band. These electrons provide a very small conductivity for pure silicon; this is referred to as the *intrinsic conductivity*. For diamond, the gap between the valence and conducting band is 7 eV, which is so large that essentially no electrons are found in the conduction band at room temperature, and thus diamond is an insulator.

The energy needed for transferring valence electrons to the conduction band can be supplied by nuclear radiation. The average energy needed to produce an electron-hole pair in silicon at room temperature is not 1.115 eV but 3.62 eV because some energy is lost as crystal excitation (at 80 K the exitation energy is 3.72 eV in Si and 2.95 eV in Ge). The electron removed from the valence band leaves a vacancy or "hole". The ionization is said to give an *electron-hole pair*. Just as the electrons move towards the anode, the holes move towards the anode.

Si has 4 valence electrons while P has 5 and In 3. If we introduce a very small amount of phosphorus into silicon, the phosphorus atoms substitute for silicon in the crystal

lattice. Each such phosphorus has an excess of 1 electron. These electrons are not free but are very weakly bound, such that only 0.04 eV is needed to transfer them into the conduction bands. Because phosphorus donates extra electrons to the system, it is referred to as a *donor material*. Silicon which contains small amounts of donor material (usually referred to as "impurity") is called *n-type silicon* since it has excess negative charge and conduction is by electrons.

If instead, indium is the impurity in the silicon crystal structure, the opposite effect is produced. Such material contains a number of energy levels only 0.06 eV above the valence band; the result is holes in the valence bands. Such material is referred to as *acceptor material*. Silicon with acceptor material is called *p-type silicon*, since the holes are considered to be positively charged. Conduction is in this case by movement of holes. The addition of controlled amounts of impurity atoms thus provide *charge carriers* (as the electrons and holes collectively are called, c.f. section 9.2) and produces the desired properties in semiconductor materials.

The most interesting effect comes from the combination of two types of silicon, one n-type and another p-type (Fig. 9.12(b)). The contact surface is referred to as a *p-n junction*. At such a junction some positive holes move to the n-type material, and vice versa. As a result a "depletion" layer a few microns thick is established at the junction where all the holes are filled with electrons and the layer is depleted of charge carriers. A p-n junction can be produced in a single piece of silicon by doping it with the proper impurity from either side of the crystal and by other techniques like ion implantation.

If a voltage is applied over the junction by connecting the negative terminal to the p-type region and the positive terminal to the n-type region, the junction is said to be *reversely biased*. With such reverse bias, the barrier height and depletion layer thickness increase. As a result the crystal opposes any current as the resistivity becomes very high. In the reverse direction the semiconductor represents a high resistance shunted by a capacitive component (Fig. 9.3(a)) due to the dielectric of the barrier layer (the p-n junction diode).

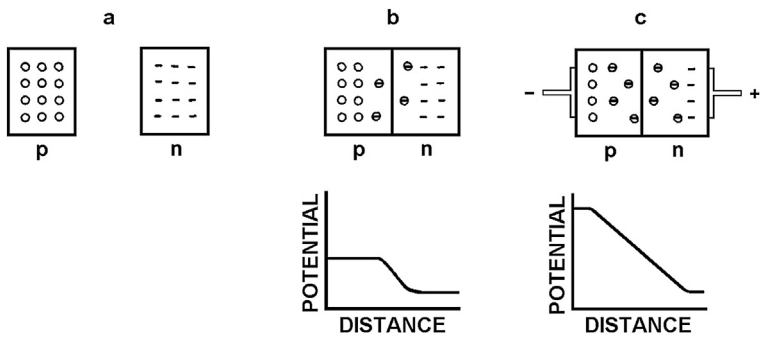


Figure 9.12 Formation and properties of a p-n junction in a semiconductor.

A number or variations of this basic design exist. The point of importance for semiconductors as nuclear detectors is that a depleted layer with a high space charge is formed. A nuclear particle entering this volume forms electron-hole pairs, which are rapidly and efficiently collected at the electrodes due to the high potential gradient. By this a charge is transported through the crystal while the original conditions are restored.

There are some formal similarities between the function of a semiconductor detector and a parallel plate ion chamber. In comparison with ionization chambers the semiconductor detector (i) requires only 3.62 eV (Si, 300 K) or 2.95 eV (Ge, 80 K) for an electron-hole pair (as compared to about 15—35 eV for an ion pair in a gas), (ii) collects the charge much faster (no slow positive ions), (iii) has a much higher stopping power, (iv) but does not normally have the property of charge multiplication. The charge through the semiconductor detector is given by Eqn (9.2) where η (the collection efficiency) is usually very close to 1, and w is 3.62 eV for silicon at 300 K and 2.95 eV for germanium at 80 K. It is seen that the signal is directly proportional to the energy absorbed in the detector as long as η is constant. Because w is much smaller in semiconductor detectors than in gas and scintillation detectors, more primary electrons are released in each absorption event, which gives better "statistics" (see section 9.11) and higher energy resolution. This makes semiconductor detectors useful for nuclear spectroscopy, although for α- and β-spectroscopy the precision is higher with magnetic deflecting devices (cf. section 3.3).

9.4.1. Surface barrier detectors

The surface barrier detector is a p-n type silicon diode wafer characterized by a rather thin depletion layer (Fig. 9.13(a)). It is made of n-type silicon on which one surface has been etched prior to coating with a thin layer of gold (typically ~40 μg/cm^2) and the other surface coated with a thin layer of aluminum (typically ~40 μg/cm^2) to provide electrical contact. This results in a window-layer which is equivalent to ~800 Å of Si. Depending on the applied voltage, the detector can be partially depleted (inactive entrance layer), totally depleted (no inactive layer), or over-depleted (higher applied potential than required for total depletion). Surface barrier detectors are used mainly for α- and β-spectroscopy and for dE/dx and E measurements for high energy particles, although the efficiency is limited by the sensitive surface diameter ($\leq$10 cm) and the energy range by depleted layer thickness ($\leq$5 mm).

The radiation sensitive depleted layer is available in various thicknesses, $\leq$ 5 mm, enough to stop electrons of $\leq$2.2 MeV, p of $\leq$32 MeV, and α of $\leq$120 MeV. A typical silicon surface barrier detector for α-spectroscopy has a sensitive area of 300 mm^2, 300 μm depletion depth, 20 keV FWHM (full width at half maximum) and operates at 100 V reverse bias. The resolving time is about 10^{-8} s. Special "rugged" detectors are

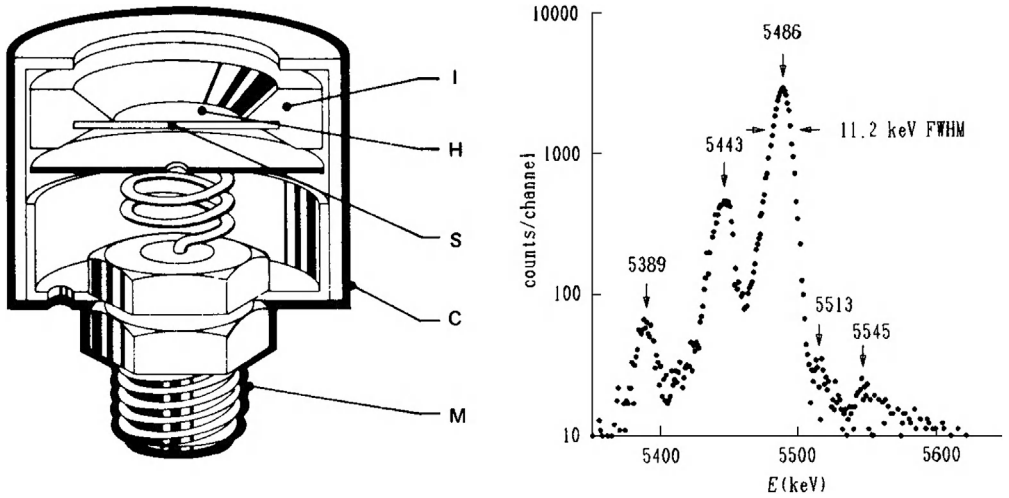


Figure 9.13 (a) Surface barrier detector; S detector wafer, H gold surface layer, M connector. (b) α-spectrum of ^{241}Am measured with a high resolution detector (Acc. to ORTEC.)

available which have an acid resistant SiO_2 surface layer permitting cleaning and contact with liquids. Detailed information for detector selection is available from various detector manufacturers.

When used for α- or β-spectroscopy, a vacuum is applied between the detector and the radiation source. In the absence of a vacuum for α-radiation the energy loss is about 1 keV per 0.001 atm per cm distance between source and detector. The absorption in the detector window for a 6 MeV α is less than 6 keV. A resolution of about 12 keV FWHM can be obtained for a 6 MeV α, Figure 9.13.

In totally depleted silicon surface barrier detectors the sensitive region extends through the whole thickness of the silicon, which may be in the form of a very thin slice (e.g. 20 μm) within two very thin conducting surface layers. A particle passing through such a detector loses a small fraction of its energy dE/dx and may then be completely stopped in a second (much thicker depleted layer) detector to lose the remainder of its energy, which may essentially be its original total E_{kin}. Particle mass A and charge Z can be determined from dE/dx and E, e.g. with the aid of the proportionality

$$E\,dE/dx \propto A\,Z^2 \tag{9.27}$$

Figure 9.14 shows a hypothetical distribution when the recorded intensities of ^{1}H, ^{2}H, and ^{3}H are plotted against $E\,dE/dx$.

As an alternative to X-ray film large plates containing a matrix of surface barrier detectors on their surface are in use. The individual detectors are pre-charged and can after exposure be individually addressed for electric readout of remaining charge in a

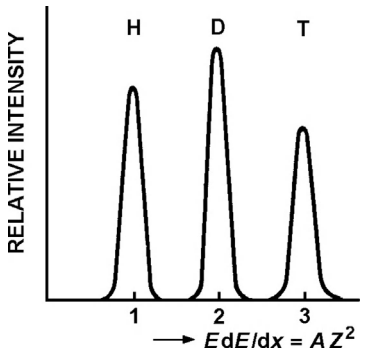


Figure 9.14 Plot of relative particle intensities for H, D, and T versus *E*d*E*/d*x* in a two-detector telescope, where the first detector records d*E*/d*x* and the second *E*.

similar way as optical MOS-sensors in digital cameras. Such plates are mainly used for X-ray imaging but also for autoradiography.

High energy particles not only cause ionization in the detector crystal but may displace some detector atoms from the crystal lattice. Radiation damage decreases with applied bias and increases with the particle mass. Such radiation damage to the crystals limits the lifetime of the detectors. The threshold dose (in particles/cm^2) is about 10^8 for fission fragments, 10^9 for α, 10^{10} for p^+, 10^{12} for fast neutrons, and 10^{13} for e^-. Radiation damage can usually be removed if the detector can be annealed at 200°C.

9.4.2. Lithium-drifted detectors

The probability of γ-interaction is so small in the small depletion depth of the surface barrier detectors that they are not very useful for γ-spectroscopy. Large depleted volumes can be created by drifting lithium atoms into a silicon or germanium crystal. Lithium does not occupy a crystal site in the crystal, but is small enough to go into interstitial sites. The ease of ionization of Li to Li^+ makes it a donor impurity. The lithium is drifted from one side of the crystal using an electric field. Its concentration at the "entrance" side becomes high and then decreases towards the other end of the crystal. The amount of lithium in the crystal adjusts itself during the drifting process so that the lithium atoms compensate the impurities. This is, however, not a stable situation when the drifting operation is stopped. Hence, the final state is "frozen" by cooling of the drifted crystal to liquid nitrogen temperature. Accidental heat-up will destroy the lithium compensation and the detector must be redrifted.

When a potential is applied over such a crystal, with the positive terminal at the high lithium side, three volumes are created, one of p-type, a middle "intrinsic" one, and an n-type one (p-i-n detectors). In the intrinsic volume the lithium donor electrons neutralize any original impurities, which are of acceptor p-type. The intrinsic volume

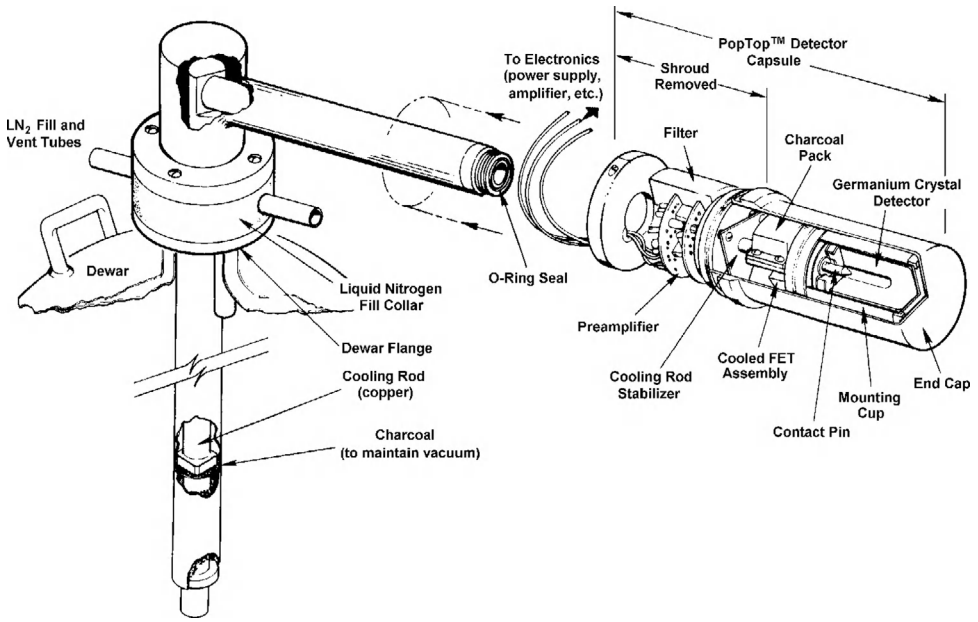


Figure 9.15 Cut-away view of a Ge-detector showing Dewar, cold-finger, preamplifier and germanium crystal. (Acc. to ORTEC.)

becomes depleted and thereby sensitive to nuclear radiation, and detectors with depleted volumes up to more than 100 cm^3 are commercially available. Figure 9.15 shows the arrangement of the Dewar vessel with liquid N_2, cold-finger, detector, and preamplifier.

Lithium-drifted detectors are made either from silicon (Si(Li) detectors) or from germanium (Ge(Li) detectors). The latter has a higher atomic number and density than silicon and is therefore preferable for γ-spectrometry. For 60 keV X-rays, the efficiency of a Si(Li) detector may be 5%, while for a comparable Ge(Li) detector it may be 100%. At lower energy the Si(Li) detector is preferable, especially if the measurements are carried out in a high γ-background. Si(Li) detectors are of particular importance in X-ray fluorescence analysis (cf. section 7.8.4).

Both types give excellent resolution, for a good Si(Li) X-ray detector about 160 eV at 5.9 keV and for a good Ge(Li) detector about 1.75 keV FWHM for 1.33 MeV γ. Energy resolution of Ge detectors varies with energy below 1.5 MeV according to the empirical eqn. FWHM $\approx (n^2 + kE_\gamma)^{½}$, where FWHM is the full peak width at half maximum, n the noise line width, $k = 2$, and E the γ-energy, all in eV units. This resolution is far superior to that of the scintillation detector, as is seen from Figure 9.16. On the other hand, the detection efficiency is usually smaller. However, an efficiency up to about 200% of that obtained with a $3' \times 3'$ NaI(Tl) scintillation detector for 1.33 MeV γ-rays (which is the reference for Ge-detector efficiency) has been obtained in the best designs, but typical values are 10−100%.

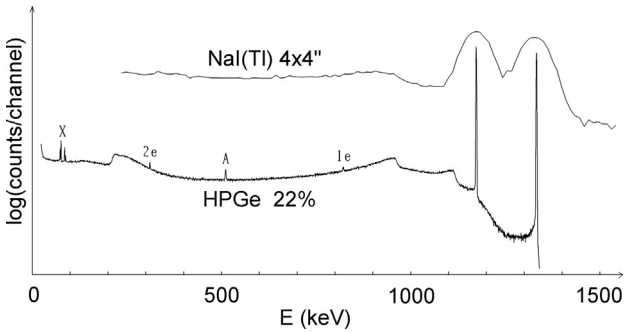


Figure 9.16 Energy spectrum of ^{60}Co obtained with scintillation and HPGe detectors.

9.4.3. Intrinsic detectors

One reason for drifting lithium into silicon and germanium is the necessity to compensate some p-type (acceptor) impurities normally present in pure materials. Detectors without lithium drifting are called *intrinsic detectors*. It is possible to increase the purity of germanium to 1:10^{13} (compared to earlier 1:10^{11}) which makes lithium-drifting unnecessary. Such material is called high purity germanium or HPGe. Detectors made of HPGe make uninterrupted cooling less important; it is not required when storing the crystals, but should be used when measuring in order to improve resolution and prevent crystal overheating. Depending on the type of dominating impurity, HPGe crystals can be of either n- or p-type. HPGe-detectors of p-type have usually a lower energy cutoff, beginning at about 100 keV, caused by the detector capsule wall and insensitive entrance layer. Commercially available p-type detectors have relative efficiencies of 10—100%. On the other hand n-type HPGe-detectors can be made with a very thin entrance layer and, when fitted with a Be-window, have a low energy cutoff beginning at about 6 keV. Relative efficiencies for n-type HPGe detectors are in general somewhat less than for p-type detectors, i.e. 10—80%. Intrinsic germanium detectors are to a large extent replacing the lithium-drifted germanium detectors because of their greater handling simplicity at no higher cost or loss of resolution.

9.5. SCINTILLATION DETECTORS

In 1908 Rutherford and Geiger established the reliability of a method of counting α-particles by observing visually the flashes of luminescence[1] produced in a thin layer of ZnS by the α-particles. Since the development of reliable *photomultiplier* tubes (PMT) in 1946, scintillating counting techniques have played an important role in nuclear science. A scintillation detector consists of a *scintillator*, or *phosphor* optically coupled to a PMT

[1] *Luminescence* includes both fluorescence and phosphorescence (sections 7.5 and 7.8).